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Task 31 : Create different types of statistical graphs.

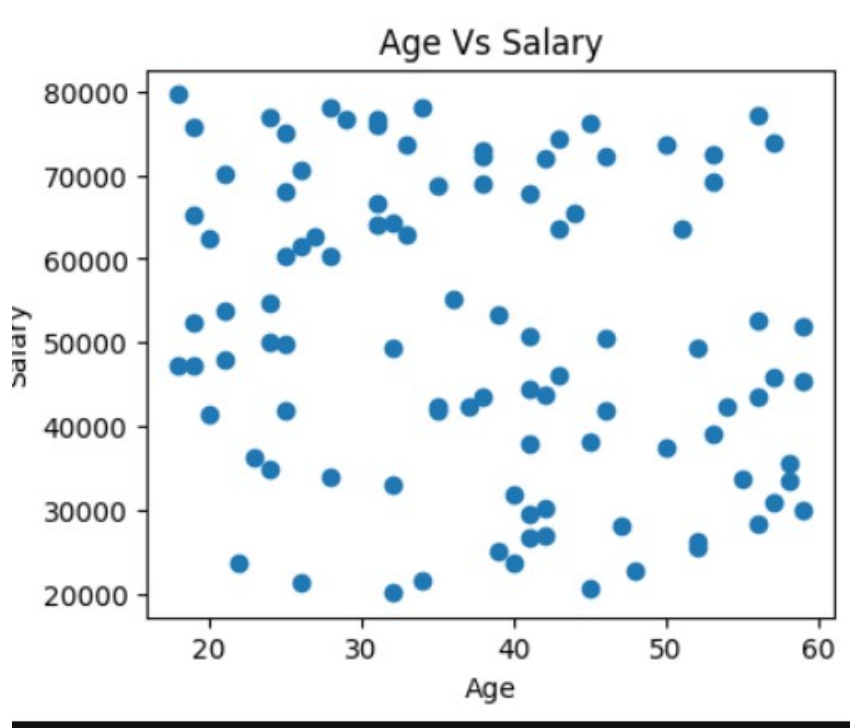
Objective

The objective of this task is to **visualize data using various statistical graphs** in order to better understand patterns, trends, distributions, and relationships within the dataset. By using graphs such as **line charts, histograms, bar charts, and scatter plots**, we can convert raw data into meaningful visual insights that are easy to interpret.

```
# Create different types of statistical graphs.

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

df=pd.read_csv("statistical_dataset.csv")
plt.figure(figsize=(10,8))
plt.subplot(2,2,1)
plt.scatter(df['Age'],df['Salary'])
plt.xlabel("Age")
plt.ylabel("Salary")
plt.title("Age Vs Salary")
plt.show()
```



I imported the required libraries such as Pandas, Matplotlib, and NumPy and loaded the dataset using the `read_csv()` function. I created a figure with a specific size and used the subplot function to organize multiple graphs in a single image. Then, I plotted a scatter graph to show the relationship between Age and Salary. I added labels to the x-axis and y-axis along with a proper title to make the graph more understandable, and finally displayed the output using the `show()` function.

I learned how to read data from a CSV file and visualize it using Matplotlib. I understood how scatter plots help in identifying relationships between variables like Age and Salary. I also learned how to use subplots to display multiple graphs in one figure and the importance of labeling and titles for better clarity. Overall, I learned that data visualization makes it easier to analyze and interpret data effectively.

Task 32 : Analyze data trends.

Objective

The objective of this task is to analyze data trends by visualizing the relationship between variables in the dataset. This helps in understanding underlying trends and making better data-driven interpretations.

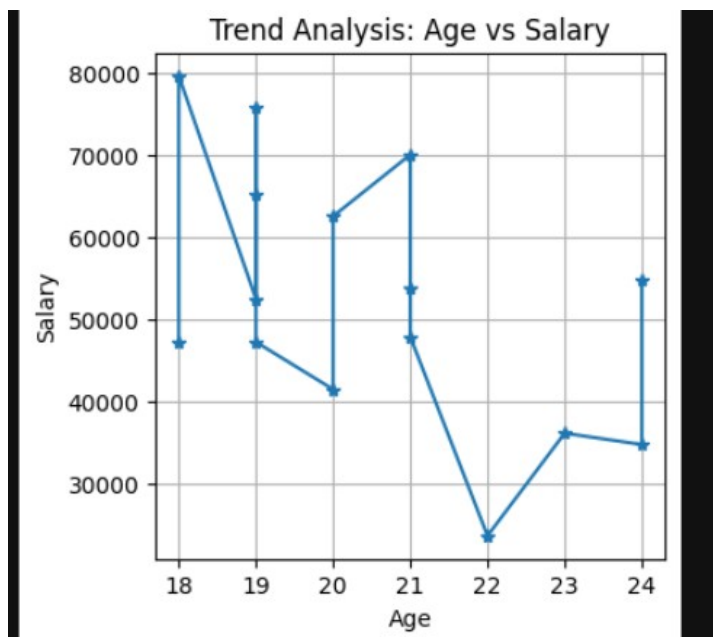
```

: df=pd.read_csv("statistical_dataset.csv")
df=df.sort_values(by="Age")
plt.figure(figsize=(4,4))
plt.plot(df['Age'].head(15), df['Salary'].head(15),marker='*')

plt.title("Trend Analysis: Age vs Salary")
plt.xlabel("Age")
plt.ylabel("Salary")

plt.grid()
plt.show()

```



I loaded the dataset, sorted it by Age, and plotted the first 15 values of Age vs Salary using a line graph with markers. I added a title, labels, and grid for better clarity.

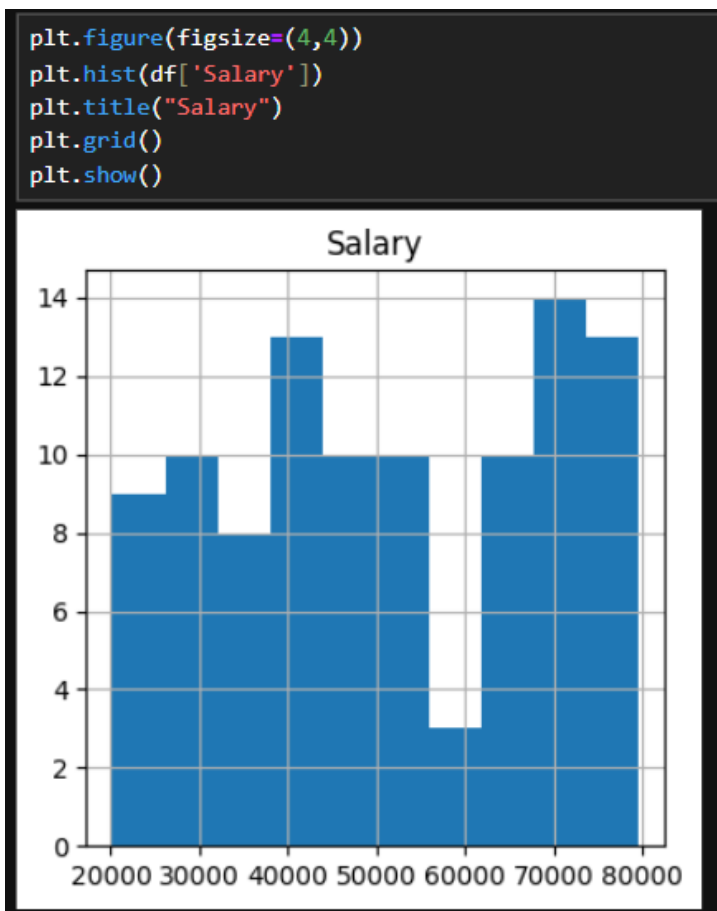
I learned that sorting data improves trend visualization and that line graphs help in understanding how one variable changes with another.

By using only a subset of data, I can focus on a smaller and clearer view of the trend. I also understood how line plots with markers help in visualizing changes between variables. This task helped me understand how trends can be identified and interpreted easily using graphs, making data analysis more effective.

Task 33 : Analyze data distribution using Histogram.

Objective:

The objective of this task is to analyze how data values are distributed using a histogram. It helps in understanding the frequency, spread, and pattern of the data, such as whether it is evenly distributed, skewed, or concentrated in certain ranges, making it easier to interpret the overall structure of the dataset.



I created a histogram using Matplotlib to analyze the distribution of the Salary column in the dataset. I used `plt.hist()` to group salary values into different ranges (bins) and visualize how frequently each range occurs.

I added a title to clearly describe the graph and displayed it using `plt.show()`, which helped in understanding how the salary data is spread across the dataset.

Task 34 : Study relationships between variables (correlation, covariance).

Objective

The objective of this task is to study the relationship between variables in a dataset using correlation and covariance. This helps in understanding how variables move together, whether they have a positive, negative, or no relationship, and how strongly they are related.

```
#Study relationships between variables (correlation, covariance).
print("Correlation : ", df.corr(numeric_only=True))
print("\nCovariance : \n", df.cov(numeric_only=True))
```

Correlation :	User_ID	Age	Salary	Experience_Years	Purchased
User_ID	1.000000	-0.004060	-0.069486	-0.083243	-0.069983
Age	-0.004060	1.000000	-0.233618	0.015190	0.158631
Salary	-0.069486	-0.233618	1.000000	-0.102595	-0.056928
Experience_Years	-0.083243	0.015190	-0.102595	1.000000	0.140126
Purchased	-0.069983	0.158631	-0.056928	0.140126	1.000000

Covariance :	User_ID	Age	Salary	Experience_Years	\
User_ID	841.666667	-1.439394	-3.608049e+04	-15.580808	
Age	-1.439394	149.315051	-5.109320e+04	1.197475	
Salary	-36080.489899	-51093.202121	3.203392e+08	-11846.792424	
Experience_Years	-15.580808	1.197475	-1.184679e+04	41.623737	
Purchased	-1.015152	0.969192	-5.094449e+02	0.452020	

	Purchased
User_ID	-1.015152
Age	0.969192
Salary	-509.444949
Experience_Years	0.452020
Purchased	0.250000

I calculated the correlation and covariance of the dataset using Pandas functions. I used `df.corr()` to measure the strength and direction of relationships between numerical variables and `df.cov()` to understand how the variables vary together.

I displayed both results to analyze the relationships between different features in the dataset.

I printed both outputs in a structured format to clearly compare the relationships between different features like Age, Salary, and Experience. This helped me analyze whether variables are positively related, negatively related, or not related at all.

Task 35 : Analyze binary data (0s and 1s).

Objective

The objective of this task is to analyze binary data (0s and 1s) and understand its distribution.

```
# Analyze binary data (0s and 1s).  
print("Purchased : ",df['Purchased'].value_counts())
```

```
Purchased : Purchased  
1      55  
0      45  
Name: count, dtype: int64
```

I calculated the frequency of values in the Purchased column using value_counts() to understand how many users purchased (1) and did not purchase (0).

Through this, I learned how to analyze binary data by quickly identifying the distribution of categories and interpreting which outcome is more common in the dataset.

Task 36 : Examine data variables (categorical, continuous) .

Objective

The objective is to identify and classify different types of variables in the dataset to understand their nature and choose appropriate analysis techniques.

```
# Examine data variables (categorical, continuous).  
print("Data Types : ",df.dtypes)
```

```
Data Types : User_ID      int64  
Age           int64  
Salary        int64  
Department    object  
Experience_Years int64  
Purchased     int64  
dtype: object
```

I used `df.dtypes` to check the data types of all columns in the dataset. This helped me identify which variables are categorical and which are continuous or numerical.

I learned how to quickly understand the structure of a dataset by examining data types, which helps in choosing the correct analysis and visualization techniques for different variables.

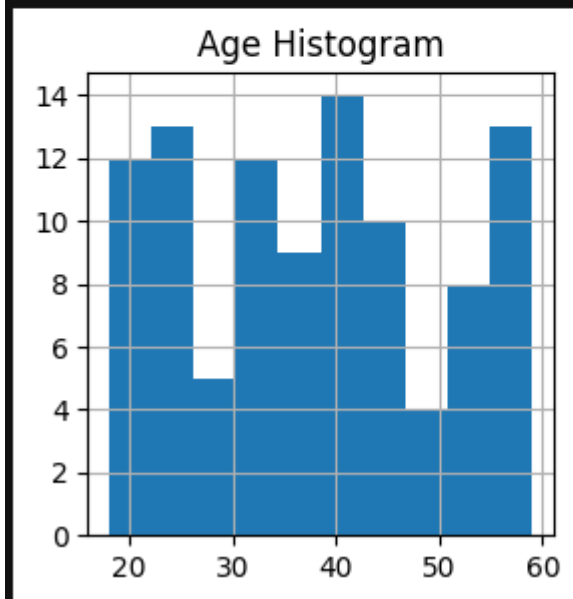
Task 37 : Visualize distribution using Histogram and Bar Charts .

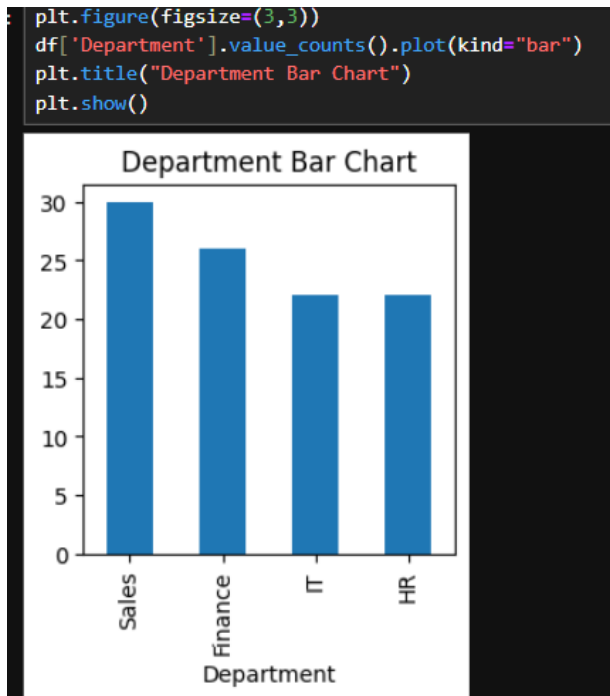
Objective

The objective is to represent data visually to understand its distribution, frequency, and comparison between categories.

```
# Visualize distribution using Histogram and Bar Charts

plt.figure(figsize=(3,3))
df['Age'].hist()
plt.title("Age Histogram")
plt.show()
```





I created a histogram to visualize the distribution of the Age column and a bar chart to show the frequency of each category in the Department column. I set the figure size, added titles to both graphs, and displayed them using Matplotlib.

I learned how to visualize continuous data using histograms and categorical data using bar charts. This helped me understand the distribution of numerical values and the comparison of categories in the dataset.

Task 38 : Visualize dataset using Pair Plot .

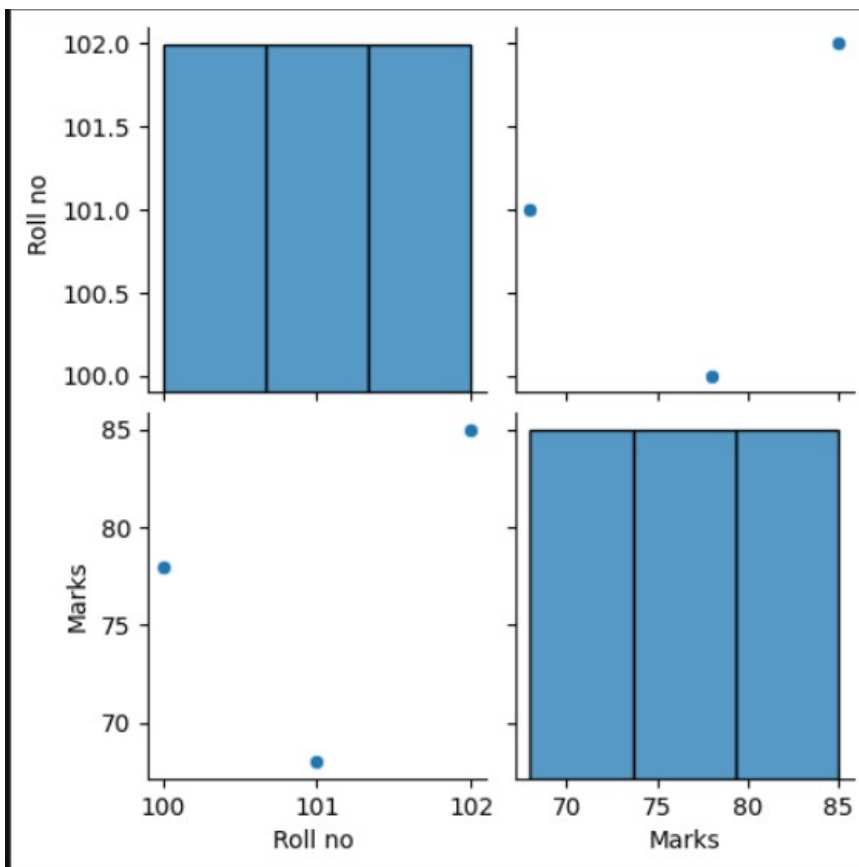
Objective

The objective of this task is to visualize the dataset using a pair plot in order to understand relationships between multiple variables at once.

It helps in identifying patterns, trends, and correlations between numerical features, as well as observing the distribution of each variable in the dataset.

```
import seaborn as sns

data={'Roll no ': [100, 101, 102],
      'Marks': [78, 68, 85]}
df=pd.DataFrame(data)
sns.pairplot(df)
plt.show()
```



I created a small dataset using a dictionary and converted it into a DataFrame using Pandas. Then, I used Seaborn's pairplot() function to visualize the relationships between the variables in the dataset and displayed the output using plt.show().

I learned how to use pair plots to visualize relationships between variables in a dataset. I understood that pair plots show both the distribution of individual variables and their pairwise relationships, which helps in identifying patterns and trends easily.

Task 39 :Identify correlation using Pearson correlation.

Objective

The objective is to measure the strength and direction of linear relationships between numerical variables.

```
# 39. Pearson Correlation
df=pd.read_csv("statistical_dataset.csv")
df.head()
print("Pearson Correlation \n: ",df.corr(numeric_only=True,method='pearson'))
```

Pearson Correlation		User_ID	Age	Salary	Experience_Years	Purchased
User_ID	1.000000	-0.004060	-0.069486	-0.083243	-0.069983	
Age	-0.004060	1.000000	-0.233618	0.015190	0.158631	
Salary	-0.069486	-0.233618	1.000000	-0.102595	-0.056928	
Experience_Years	-0.083243	0.015190	-0.102595	1.000000	0.140126	
Purchased	-0.069983	0.158631	-0.056928	0.140126	1.000000	

I loaded the dataset using Pandas and displayed the first few rows using df.head() to understand its structure. Then, I calculated the Pearson correlation between all numerical variables using df.corr() with the Pearson method and printed the result to analyze relationships in the dataset.

I learned how to use Pearson correlation to measure the strength and direction of relationships between numerical variables. I also understood how to quickly explore a dataset and identify how different features are related to each other.

Task 40 : Check consistency of data distribution .

Objective

The objective is to analyze the shape and spread of data using measures like skewness and kurtosis to determine whether the data is normally distributed or not .

```
# 40. Distribution Consistency

print("Skewness : \n",df.skew(numeric_only=True))
print("Kurtosis : \n",df.kurt(numeric_only=True))

Skewness :
  User_ID      0.000000
  Age        0.100556
  Salary     -0.048454
  Experience_Years  0.064064
  Purchased  -0.204082
dtype: float64
Kurtosis :
  User_ID      -1.200000
  Age        -1.123502
  Salary     -1.282005
  Experience_Years -1.492236
  Purchased  -1.998738
dtype: float64
```

I calculated the skewness and kurtosis of the numerical columns in the dataset using Pandas functions. I used `df.skew()` to measure the asymmetry of the data distribution and `df.kurt()` to understand the peakedness and presence of outliers, and printed both results for analysis.

I learned how to analyze the shape and consistency of data distribution using skewness and kurtosis. I understood how skewness indicates whether the data is left or right skewed, and how kurtosis helps identify whether the data has extreme values or follows a normal distribution.